ASEN 5090 HW1

Assigned: 26 August 2026

Due: 4 September 2026

1. (M&E Book Homework Problem 1-4)

A *pseudolite* (short for pseudo-satellite) consists of a generator of a GPS-like signal and a transmitter. Pseudolites are used to augment the GPS signals. Suppose an observer is constrained to be on the line joining two pseudolites PL1 and PL2, which are separated by 1 km. The pseudolite clocks are perfectly synchronized but the observer's clock may have an unknown bias with respect to the pseudolite clocks. Estimate the observer's position and clock bias given that the pseudoranges to PL1 and PL2 are (a) 550 m and 500 m, respectively, and (b) 400 m and 1400 m, respectively.

2. Measurement Geometry

A ground station located at x_G=0, observes an aircraft flying by at constant speed (v0) at constant altitude (h0), traveling straight from horizontal position –x0 to +x0, passing directly above the ground station location. The ground station is able to make measurements of the zenith angle (z), range (R), and range rate ($\dot{R}$) to the aircraft. We are interested in studying how useful these measurements are for determining the x-position of the aircraft (x_A).

a) Derive expressions for each of the measurements - range, range rate, and zenith, as a function of x_A , and the parameters v0 and h0.
b) Write a simple, well-organized program/code to simulate the aircraft motion for specified values of v0, x0, and h0; and for this motion, compute and plot the range, range rate, and zenith angle measurements based on the expressions you found in part a. Plot the measurements for the following values: v0=50m/s, x0=250m, h0=100 m.

Now, assume that the aircraft height (h0) is known perfectly, and the measurements are to be used to estimate the horizontal position of the aircraft, x_A.

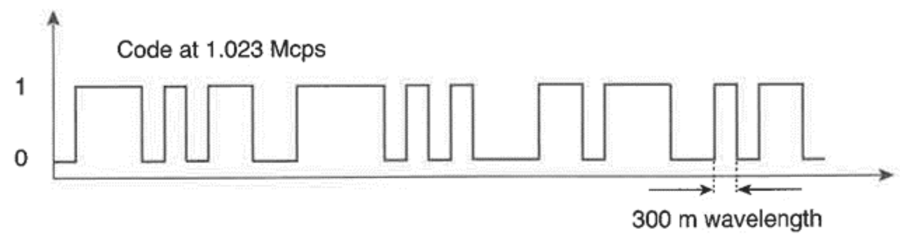
c) How useful are the three types of measurements for determining x_A? In other words, how sensitive is each measurement to the aircraft position? For which portion(s) of the flight does each measurement work well and where is it not very useful? Explain.
d) Does the sensitivity/utility of the observations get better if the aircraft is flying at a lower altitude? Explain.
e) (Optional – How would an error in the assumed height affect your results?)

3. M&E Book Problem 2-1 See next page

4. M&E Book Problem 2-2

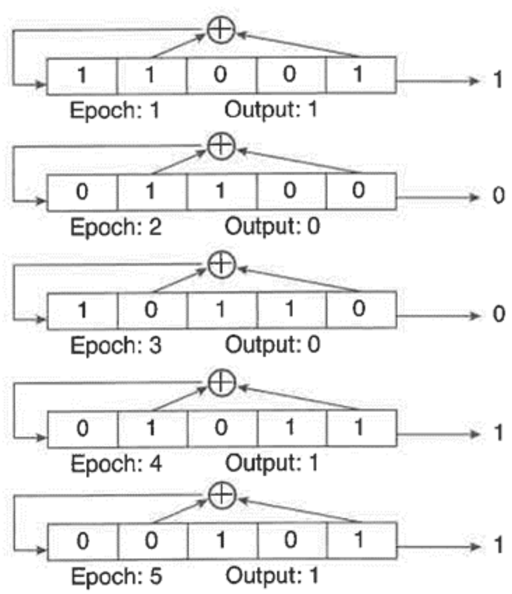
Homework Problems

2-1. A C/A-code is a series of binary *chips* that may look like:



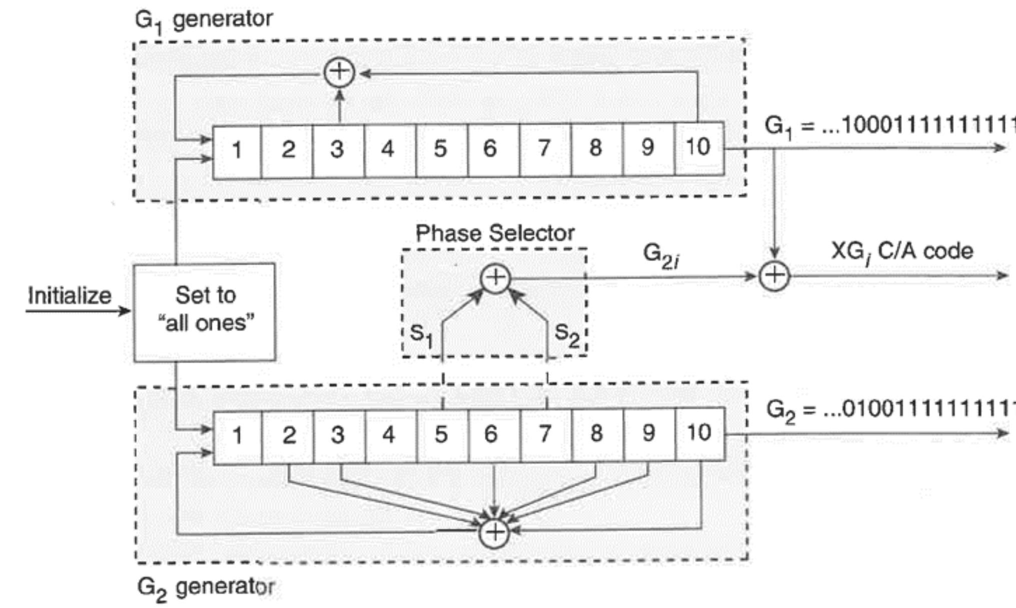
The C/A-codes can be generated with *bit shift registers*. Bit shift registers are arrays of bits (1's and 0's) of specified length. At each clock cycle (or *epoch*), all of the bits are moved one space to the right in the array. The rightmost bit is output, and a new leftmost bit is generated by an operation on the bits from the previous clock cycle.

The following is an illustration of a 5-bit long shift register in action.



Thus, the output sequence from this register, given that this register was initialized with 11001, is $\{1, 0, 0, 1, 1, \ldots\}$. Note that the sequence can go on forever, but eventually the output will repeat. The $\oplus$ symbol indicates "modulo 2 addition" (or "bitwise XOR"): $0 \oplus 0 = 0$; $0 \oplus 1 = 1$; $1 \oplus 0 = 1$; $1 \oplus 1 = 0$. You can add more than two bits: $1 \oplus 1 \oplus 1 = 1$; $1 \oplus 1 \oplus 1 \oplus 1 = 0$; etc.

The satellites are identified by their C/A-code or pseudorandom noise (PRN) sequence number. We often use 'PRN *k*' to identify both a C/A-code and the satellite transmitting it. To create the C/A-codes, GPS satellites use two 10-bit shift registers in the configuration shown below.



Note the phase selector in the middle of the diagram. S_1 and S_2 indicate which bits of the G_2 shift register are added to create the G_{2i} output at each epoch. S_1 and S_2 are different for the different satellites. For example, PRN 1 is generated by adding bits 2 and 6 from the G_2 shift register to form the G_{2i} bit; PRN 2 is formed by adding bits 3 and 7. The C/A-code output bit is XG_i. For code selection, see (CD) *Documents\IS-GPS-200D.pdf*, Table 3-I, pp. 7–8.

Briefly, the algorithm for creating the C/A-codes is:

i. Load both the G_1 and G_2 shift registers with 'all 1s'.
ii. Compute the sums from all the $\oplus$ operations to determine the output bit for the current epoch.
iii. Shift both registers one element to the right; load the leftmost elements of G_1 and G_2 with the appropriately calculated bits from just prior to the shift.
iv. Go back to step (ii).

(MATLAB Hint: Let **G** be a 1×10 matrix representing a shift register. A quick way to shift the register by one position is as follows: `G = [newBit G(1:9)]`, where `newBit` is the new leftmost element.)

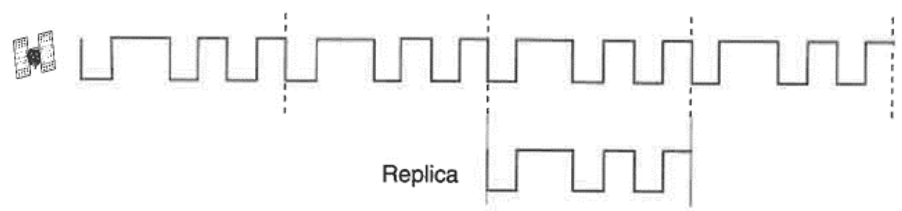
(a) Create the entire 1023-chip C/A-code for PRN 19 as a vector in MATLAB (or any other math program you'd like to use). Plot the first 16 and last 16 chips of the code. It'll be easier for you and your instructor to check the answer if you express it in hexadecimal. Verify that the first 16 chips of PRN 19, expressed in hexadecimal, are E6D6.

(b) Create the C/A-code output for PRN 19 from epochs 1024 to 2046 as a 1023-element vector. How does this vector compare with the one you generated in part (a)?

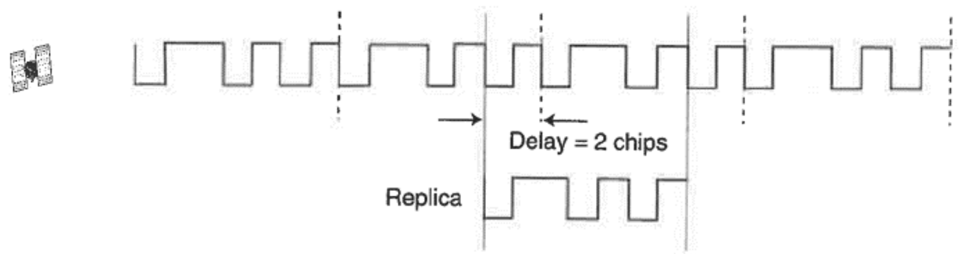
(c) Repeat part (a) for PRN 25 (we will use this C/A-code later).

(d) Repeat part (a) for PRN 5 (this C/A-code will also be used later).

2-2. A GPS receiver measures pseudorange to a satellite by examining the correlation of the received signal with a replica generated by the receiver. The receiver compares its C/A-code replica with a continuously repeating C/A-code from the satellite, as shown in the following figure, where the signal is "frozen" in time.



If the user moves away from the satellite, the repeating C/A-code's first bit has its arrival time effectively delayed with respect to the C/A-code replica, as shown below:



You will need code from Problem 2-1 to solve this problem. However, you should first convert the PRN sequences from $\{0,1\}$ to $\{1,-1\}$: $0 \to 1, 1 \to -1$. This change is required because we want the non-matching parts of the PRN sequences to subtract from the result in the correlation process. (MATLAB Hint: You can transform your PRN sequences from Problem 2-1 as follows: `CA_Problem2= (CA_Problem1==0)*(1) + (CA_Problem1==1)*(-1);`)

Denoting the C/A-codes (or PRNs) associated with satellites k and l as $x^{(k)}(i)$ and $x^{(l)}(i)$, respectively, $i = 0, 1, 2, \ldots, 1022$, let us define their auto- and cross-correlation functions as follows. The normalized auto-correlation function of PRN k for shift n ($= 0, \pm 1, \pm 2, \ldots$) is defined as

$$R^{(k)}(n) = \frac{1}{1023} \sum_{i=0}^{1022} x^{(k)}(i) x^{(k)}(i+n)$$

and the normalized cross-correlation between PRNs k and l is defined for shift n as

$$R^{(k,l)}(n) = \frac{1}{1023} \sum_{i=0}^{1022} x^{(k)}(i) x^{(l)}(i+n)$$

where $x^{(\bullet)}(m + 1023) = x^{(\bullet)}(m)$. Note that both correlation functions are periodic, and repeat as the shift exceeds 1023. (MATLAB Hint: A shifted version of a PRN can be created as follows: `CA_Replica = CA_Replica([end (1:end - 1)]);`)

(a) Plot $R^{(19)}(n)$, the auto-correlation function of PRN 19.

(b) Create a 1023-chip PRN sequence as PRN 19 delayed by 200 chips. Plot the normalized cyclic cross correlation function of PRN 19 with this delayed version. Is the peak of the correlation where you had expected it to be?

(c) Plot the cyclic cross-correlation function of PRN 19 and PRN 25. How does this plot compare with the one you made in part (a)?

(d) Plot the cyclic cross-correlation function of PRN 19 and PRN 5. How does this plot compare with the one you made in part (c)?

(e) Create three 1023-chip PRNs as follows. Define x_1 as PRN 19 delayed by 350 chips; x_2 is defined as PRN 25 delayed by 905 chips; and x_3 is PRN 5 delayed by 75 chips. Sum these three PRNs together and correlate the result with a replica of PRN 19. Is the peak of the correlation where you had expected it to be? (Hint: Addition here means simple addition: $1 + 1 = 2$, and $1 - 1 = 0$.)

(f) In order to simulate the effect of radio frequency noise on acquisition and tracking of a C/A-code, create a 1023-element vector *noise*, each element of which is normally distributed with zero mean and standard deviation of 4. Plot x_1, x_2, x_3, and *noise* in separate plots on a page. Scale the vertical axes of these plots so that they all have identical ranges. (MATLAB Hints: (1) `noise = 4*randn(1,1023);` (2) Use the command *`subplot`* to generate multiple plots on a page.)

(g) Now sum x_1, x_2, x_3, and *noise*, and correlate the result with a replica of PRN 19. Is there a correlation peak that stands out from the others? Is it where you'd expect it to be? Are you surprised by the ability of PRN 19 to survive the interference from *noise* and the other PRNs?

References

Borre, Kai *et al.* (2007). *A Software-Defined GPS and Galileo Receiver: A Single-Frequency Approach*, Birkhäuser.

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